DES646: Nuvé - An AI-Based Fashion Trend Forecasting System Using Interactive Data Visualization

Team Blue Popcorn — Technical Paper

Members:

Name	**Roll No.**	**Email**
Anushree Mittal	230178	anushreem23@iitk.ac.in
Ayush Raj	230266	ayushraj23@iitk.ac.in
Ananya Verma	230136	ananyav23@iitk.ac.in
Hemesh Saini	230468	hemeshs23@iitk.ac.in
Ivy Surwade	230490	ivysanjay23@iitk.ac.in

Abstract

The fashion industry is undergoing a paradigm shift driven by data-centric innovation. Predicting future fashion trends no longer relies solely on intuition or qualitative observation but increasingly leverages artificial intelligence and large-scale data analysis. This paper presents *Nuvé*, an AI-driven platform designed to forecast emerging fashion trends through multimodal data—visual imagery, text, and sentiment analytics. The system combines machine learning models, including YOLO for image segmentation, PCA and autoencoders for feature extraction, K-Means clustering for style grouping, and AdaBoost for sentiment analysis. Integrated through an interactive Streamlit dashboard, *Nuvé* visualizes fashion evolution across attributes such as color, texture, and silhouette. The project is developed under the DES646 mid-term phase, emphasizing methodological

design, model integration, and dashboard prototyping for real-time visualization.

Keywords: Fashion forecasting, Artificial intelligence, YOLO, SAM, PCA, K-Means, AdaBoost, Streamlit, Data visualization, Trend analysis

1. Introduction

The rapid digitization of the fashion ecosystem has redefined how trends originate, evolve, and are consumed. Social-media platforms, e-commerce interfaces, and digital fashion repositories now serve as dynamic sources of behavioral and aesthetic data. Designers and analysts can leverage this data to anticipate cultural shifts, streamline design decisions, and respond to evolving consumer preferences.

Nuvé: An AI-Driven Fashion Trend Forecasting and Visualization System explores the intersection of artificial intelligence, machine learning, and visual analytics in predicting the future of style. The goal is to create a functional, interactive system that can identify early indicators of trend formation using real-world data and present these insights through an intuitive, visual interface. The project, carried out by **Team Blue Popcorn** as part of **DES646**, aims to contribute to data-informed design research and creative forecasting.

2. Related Work

Existing literature in computational fashion forecasting has explored various approaches to automate trend prediction. Early work emphasized visual learning through CNNs for garment recognition [3], while others focused on text mining from fashion blogs and consumer reviews [1]. More recent research integrates multimodal approaches—combining textual sentiment with visual trend data [4].

Visualization frameworks such as Plotly, Tableau, and Streamlit have enhanced data interpretability in creative domains [2]. However, few systems unify advanced AI segmentation models like YOLO and SAM, with statistical clustering and sentiment analysis within an interactive, open-source dashboard. *Nuvé* bridges this gap by merging predictive modeling with design visualization to support both technical and creative decision-making.

3. Aim and Objectives

The primary aim of this mid-term project is to design and implement an AI-based fashion forecasting tool capable of identifying and visualizing trend trajectories. The system's objectives are:

- To collect and preprocess large-scale visual and textual fashion data.

- To apply advanced AI/ML models for feature extraction, clustering, and sentiment analysis.
- To forecast attributes such as color, texture, silhouette, and pattern evolution.
- To integrate an interactive Streamlit dashboard for trend exploration.
- To support data-driven decision-making for designers and consumers.

4. Methodology and System Architecture

The *Nuvé* system is structured around four main components: data collection, feature processing, machine-learning modeling, and visualization interface.

4.1 AI/ML Methods and Tools

- **Image Segmentation:** YOLO (You Only Look Once) and SAM (Segment Anything Model) are used to extract garment elements such as tops, dresses, or accessories from images.

Figure 1: Image Segmentation Output

- **Feature Extraction:** Autoencoders and Principal Component Analysis (PCA) reduce dimensionality while retaining essential visual characteristics.
- **Clustering:** K-Means groups similar fashion items based on extracted features, enabling the system to identify stylistic clusters.
- **Sentiment Analysis:** AdaBoost is applied to social-media comment datasets (e.g., Instagram) to evaluate audience sentiment toward particular trends.
- **Visualization:** Power BI and Streamlit are used to visualize evolving trends across time, color palettes, and sentiment shifts.

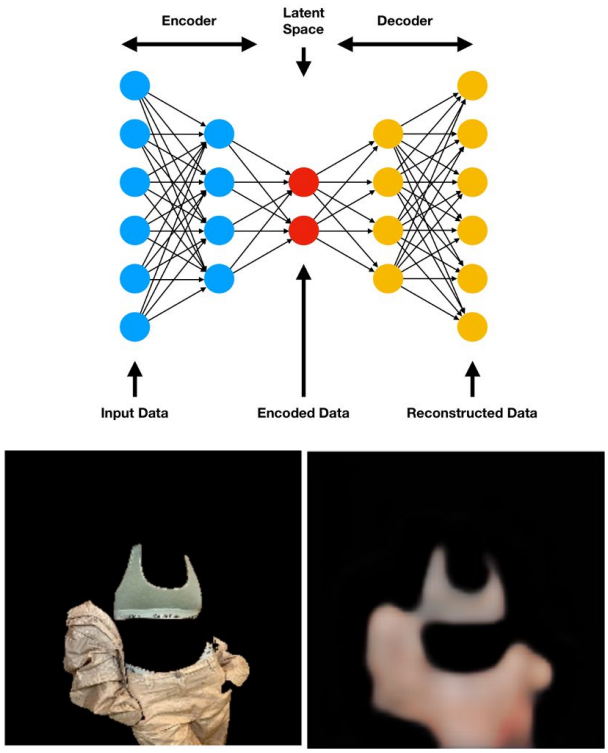

Figure 2: K-Means Clustering Visualization

4.2 Data Pipeline

- **Data Acquisition:** Collect datasets from DeepFashion, Fashion-MNIST, Kaggle, and scraped Instagram data tagged by fashion-related keywords.
- **Preprocessing:** Perform normalization, image resizing, and removal of duplicates. Text data undergoes tokenization, stop-word removal, and lemmatization.
- **Feature Extraction and Analysis:** Extract stylistic features (color histograms, texture vectors) and textual sentiments.
- **Visualization:** Integrate output into Streamlit dashboards and Power BI interfaces for interpretive analysis.

5. Design Process

Following an iterative design-research approach, the *Nuvé* development process includes:

1. User and Market Research: Understanding stakeholder needs—designers, students, and fashion analysts.
2. Data Exploration: Identifying key visual and textual features influencing trends.
3. Model Development: Training segmentation, clustering, and sentiment models.
4. Dashboard Prototyping: Building an interactive prototype for exploratory analysis.
5. Testing and Feedback: Evaluating usability and accuracy, refining user interface and model output.

6. Implementation and Visualization

The *Nuvé* dashboard, developed using Streamlit and Plotly, presents an interactive interface that allows users to:

- View real-time data summaries (trend frequency, color analysis, sentiment ratio).
- Explore fashion clusters and attribute distributions through interactive charts.
- Access modular panels for visual vs. textual analytics.

Caching mechanisms (`@st.cache_data`) ensure high-performance data retrieval. The system is designed with responsive UI components, allowing scalable deployment and future model integration.

7. Results and Discussion (Mid-Term Stage)

At the mid-term milestone, *Nuvé* has successfully established its data pipeline, model selection framework, and visualization prototype. Preliminary results indicate that YOLO and SAM can efficiently segment garments with high visual accuracy, while K-Means clustering effectively identifies stylistic groups. Sentiment analysis using AdaBoost demonstrates promising accuracy in classifying positive and negative perceptions of fashion trends.

The dashboard currently visualizes correlations between sentiment and fashion attributes, providing interpretable insights. User testing among peers highlighted the system's potential as an educational and design research tool. Upcoming development phases will focus on deeper predictive modeling and integrating generative visualization.

8. Expected Deliverables

By the project's completion, the following deliverables will be achieved:

- An interactive prototype dashboard showcasing trend forecasting outcomes.
- A curated dataset of multimodal fashion data.
- A concept video demonstrating workflow and interface design.
- Documentation covering methodology, UI/UX flow, and model evaluation.
- A draft publication prepared for design and AI research venues such as DRS 2026 or India HCI 2026.

9. Conclusion and Future Scope

The *Nuvé* system represents an innovative intersection of AI, visualization, and fashion research. By fusing segmentation, clustering, and sentiment modeling, it transforms raw

data into actionable design insight. The mid-term results validate the system's technical foundation and its applicability in creative forecasting contexts.

Future work will involve refining model accuracy, integrating generative AI for visual trend synthesis, and expanding multilingual sentiment analysis. The ultimate goal is to create an intelligent, scalable tool that empowers both designers and researchers to anticipate and shape the aesthetics of future fashion.

References

[1] Hassan, M., & Lee, S. (2018). Text mining for fashion trend analysis: From blogs to brands. *Journal of Data-Driven Design, 12*(3), 45–59.

[2] Kim, Y., & Park, D. (2022). Visual analytics in fashion forecasting using interactive dashboards. *Information Design Journal, 31*(2), 102–118.

[3] Liu, Q., Wang, J., & Chen, H. (2016). Deep learning for fashion image classification. *IEEE Transactions on Multimedia, 18*(6), 1177–1187.

[4] Yang, T., Zhou, K., & Singh, P. (2020). Multimodal trend prediction in online fashion communities. *ACM Transactions on Intelligent Systems and Technology, 11*(5), 1–20.